

EduTech Challenge Track • AppSprint 2026

Pomly

A local-first focus timer, flashcard companion, and study activity tracker designed for seamless, distraction-free learning.

The Challenge & Our Solution

The Problem

- ❗ Fragmented tools: Separate apps for timers, notes, tasks, and tracking.
- ❗ Network dependency & cloud sign-up requirements compromise data privacy.
- ❗ Tedious manual workflows for converting lecture notes into revision cards.

The Solution: Pomly

- ✅ Unified privacy-friendly ecosystem: Timer, flashcards, and todos in one app.
- ✅ 100% Local-First: Powered by SharedPreferences with no account or backend needed.
- ✅ Smart local text parsing & PDF extraction to instantly generate revision decks.

Core Features



Pomodoro Engine

Configurable focus and rest periods with automated cycle progression, leading to a calculated 5x long recovery rest.



Flashcard Engine

Import selectable PDFs or raw notes (Q&A, term :: definition) with built-in Spaced Repetition (Again, Hard, Good, Easy ratings).



Activity Tracking

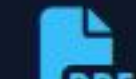
Shared sync between focus minutes, streaks, daily todos, and flexible weekly goals without losing historical progress.

Architecture & Tech Stack

Technical Framework

-  **Frontend:** Flutter & Dart with Material 3 widgets.
-  **State Management:** Provider pattern for reactive app-wide state.
-  **Persistence:** shared_preferences for offline local store.

Integrated Services

-  **file_picker & syncfusion_flutter_pdf:** Local PDF text harvesting.
-  **Local Parsing Engine:** Proprietary Dart parser for structuring study cards offline.

Clean Feature-Based Architecture

lib/features

Modular screens for cards, focus timer, streak dashboard, and navigation home screen.

AppState

Centralized state coordinator uniting timers, task completion, and daily goal logs.

Local Engine

Zero external API calls required for parsing text or evaluating flashcard mastery.

Team & Expected Impact

Team meow

 **Nandana Rajan:** Developer

 **Sandra B.A.:** Designer

 Built during AppSprint Solution Challenge 2026 (muLearn LBSITW).

Target Impact

-  Provides students with a frictionless, offline-capable study habit builder.
-  Total data privacy with zero server storage requirements.
-  Sustained study consistency through integrated goals and streak tracking.